



AI SINGULARITY: The Awakening of Machines or a Philosophical Mirage?

An Analysis of Reality from the Thomistic Perspective

Roberto I. Asplanato

June 2026

"For our knowledge is imperfect... but when the perfect comes, the imperfect will pass away."

1 Corinthians 13:9–10

Abstract

This paper analyzes the hypothesis of the technological singularity in artificial intelligence—the moment at which AI would irreversibly surpass human intelligence—from an interdisciplinary perspective integrating computer science, formal and natural sciences, neuroscience, and the Thomistic framework adopted as the logical foundation of the analysis.

The paper defines and classifies AI and projects a hypothetical timeline of its evolution. It then distinguishes between types of singularity—functional and strong ontological—focusing the analysis on the latter. The core tenets of Thomism, its key principles, and concepts are subsequently set forth. The human cognitive process is explained and contrasted with what AI can produce.

Two contributions from contemporary science that engage with this perspective are then incorporated: the contemporary cosmological framework (the Big Bang and the fine-tuning of physical constants) and Gödel's incompleteness theorem. The recent Encyclical *Magnifica Humanitas* (Leo XIV, 2026), which provides ethical and social guidance for the global governance of AI, is likewise integrated.

The result is a well-grounded conclusion regarding the singularity in artificial intelligence, demonstrating that the strong ontological singularity is philosophically incoherent within the Thomistic framework. Finally, the paper poses the true challenge of our era: what place the human being will occupy by the end of this century, in the face of a technocratic paradigm that threatens to reduce persons to mere cogs serving the interests of large supranational entities.

Keywords: artificial intelligence; technological singularity; Thomism; ontology; strong ontological singularity; human person.

1. Introduction

Does your virtual assistant feel anything when it converses with you? If you have ever interacted with a virtual assistant, you may have thought: “it seems so human.” But does it really understand what it says? Could it ever possess consciousness, emotions, or desires of its own?

This question is not merely technical; it is also philosophical and ethical. In the current debate on technological singularity, some futurists—such as Ray Kurzweil (2005)—argue that machines will surpass us entirely and that we will eventually merge with them. Others, such as Sam Altman, CEO of OpenAI, foresee a profound and gradual transformation, advocating for governance frameworks to orient AI development toward beneficial outcomes while acknowledging its potential long-term risks (Altman, 2023). From a philosophical tradition of more than 700 years—Thomism—the answer is clear: a machine, however sophisticated, will never be a person.

It should be noted from the outset that this conclusion is not unanimous in contemporary philosophy. Currents such as functionalism (Putnam, 1967; Fodor, 1975) maintain that the mind can be reproduced in any substrate that performs the same cognitive functions; computationalism (Turing, 1950; Minsky, 1967) proposes that thought is, in essence, information processing; and emergentism holds that properties such as consciousness can arise spontaneously from systemic complexity, without requiring a prior vital principle. These positions form the counterpoint against

which the Thomistic framework builds its argument. The aim of this paper is not to ignore those positions, but to demonstrate that the Thomistic framework provides more robust answers to the questions posed by technological singularity. The analysis is complemented by the recent Encyclical *Magnifica Humanitas* (Leo XIV, 2026), which contributes a magisterial perspective on human dignity in the face of AI and its ethical and social implications. The challenge, however, is not exhausted at the ontological level—whether a machine can become a person: there exists a second risk, of a different nature, that of the technocratic paradigm, which, without any machine awakening or any human merging with one, equally reduces the person—not through fusion, but through instrumentalization—to a cog serving the efficiency, profit, or control of a few. This paper addresses both fronts: the philosophical, which inquires whether a machine can become a person, and the social, which inquires what place the human being will occupy in a world of increasing hyperautomation.

This paper presents the Thomistic framework in an accessible manner, without presupposing prior philosophical training on the part of the reader, and demonstrates its implications for regulation, labor, and human coexistence.

1.1 Methodological Note

This paper adopts an interdisciplinary approach combining philosophical-deductive analysis, a review of specialized literature in computer science, mathematics, physics, and neuroscience, and an analysis of magisterial documents. The Thomistic framework is employed as a normative reference system—not as an empirical description—which means that its conclusions depend on the rational acceptance of its foundational metaphysical premises. The scope of the paper is analytical-argumentative; it does not aim to experimentally refute computationalist positions, but to demonstrate the internal coherence and explanatory robustness of the Thomistic framework in relation to them.

The term strong ontological singularity is proposed by the author of this paper to designate the most radical variant of the singularity scenario, and does not constitute standard terminology in the prior literature.

2. Artificial Intelligence (AI)

Given that no single, consensus definition of AI exists among experts, this paper opts for an original integrated definition that brings together the fields of knowledge mentioned above, with the aim of articulating a broad range of analytical perspectives.

2.1 Integrated Definition

Artificial intelligence (AI) is a field of computer science that develops computational systems, inspired by human cognition, capable of processing large volumes of data, learning from experience, adapting after deployment, and operating with varying degrees of autonomy. These systems perform operations that imitate or are equivalent to cognitive functions such as perception, reasoning, or decision-making, and generate predictions, recommendations, or content. In essence, AI functions as a multilevel automation: a system that organizes and executes tasks in successive layers, from basic data processing to complex decision-making. From the Thomistic perspective that

frames this paper, AI constitutes an artifact: a system whose form, ends, and operating principle are entirely imposed from outside by its designers, with no intrinsic principle of activity residing within it.

2.2 Classification of Artificial Intelligence

From a conceptual perspective widely accepted in the specialized literature (Russell & Norvig, 2020), artificial intelligence is classified into three levels:

- **Artificial Narrow Intelligence (ANI)**, also called “weak” or “task-specific” AI: refers to systems designed and trained to execute a specific task or a limited set of functions (e.g., facial recognition, recommendation systems, or machine translation). These systems lack consciousness, genuine semantic understanding, or emotional states.
- **Artificial General Intelligence (AGI)**: a hypothetical type of intelligence that could carry out any intellectual task that an average human being is capable of performing. Unlike narrow AI, an AGI could transfer learning across domains, reason flexibly, and adapt to novel contexts without specific retraining. To date, no functional AGI has been developed.
- **Artificial Superintelligence (ASI)**: a theoretical category proposed in future technological singularity scenarios (Bostrom, 2014). A superintelligence would vastly surpass human cognitive performance in virtually every domain. Its study belongs primarily to philosophical and prospective speculation.

Note. ANI: Artificial Narrow Intelligence; AGI: Artificial General Intelligence; ASI: Artificial Superintelligence.

2.3 Evolution and Prospects: Toward a Gradual Progression

The foregoing tripartite classification has been criticized by some authors for being excessively discrete. Recent literature proposes a progression through gradual stages, in which general intelligence would not appear disruptively but through incremental milestones. Table 1 presents a hypothetical timeline for illustrative purposes.

Table 1. Hypothetical Chronological Evolution of AI

Estimated Period	Type (Proposed Designation)	Key Characteristics
1950–2020	Reactive / Narrow AI	Specialized in a single task; no long-term memory.
2020–2026	Emerging General-Purpose AI	Foundation models (GPT, Gemini) with versatile capabilities, but without consciousness or full generalization.
2027–2030	Early AGI	Performance comparable to the average human in diverse tasks.
2030–2035	Self-Improving AI (supervised)	Capacity to redesign its own algorithms under human oversight.
2035–2040	Superintelligence	Performance vastly superior to humans in science, creativity, and strategy.
> 2045	Human-Machine Fusion and Mind Transfer	Cognitive integration (mind uploading) via Brain-Computer Interface (BCI). Highly speculative status.

Note. Dates and designations are speculative and proposed by the author based on the cited literature; they are presented solely to illustrate a possible trajectory, not as confirmed predictions.

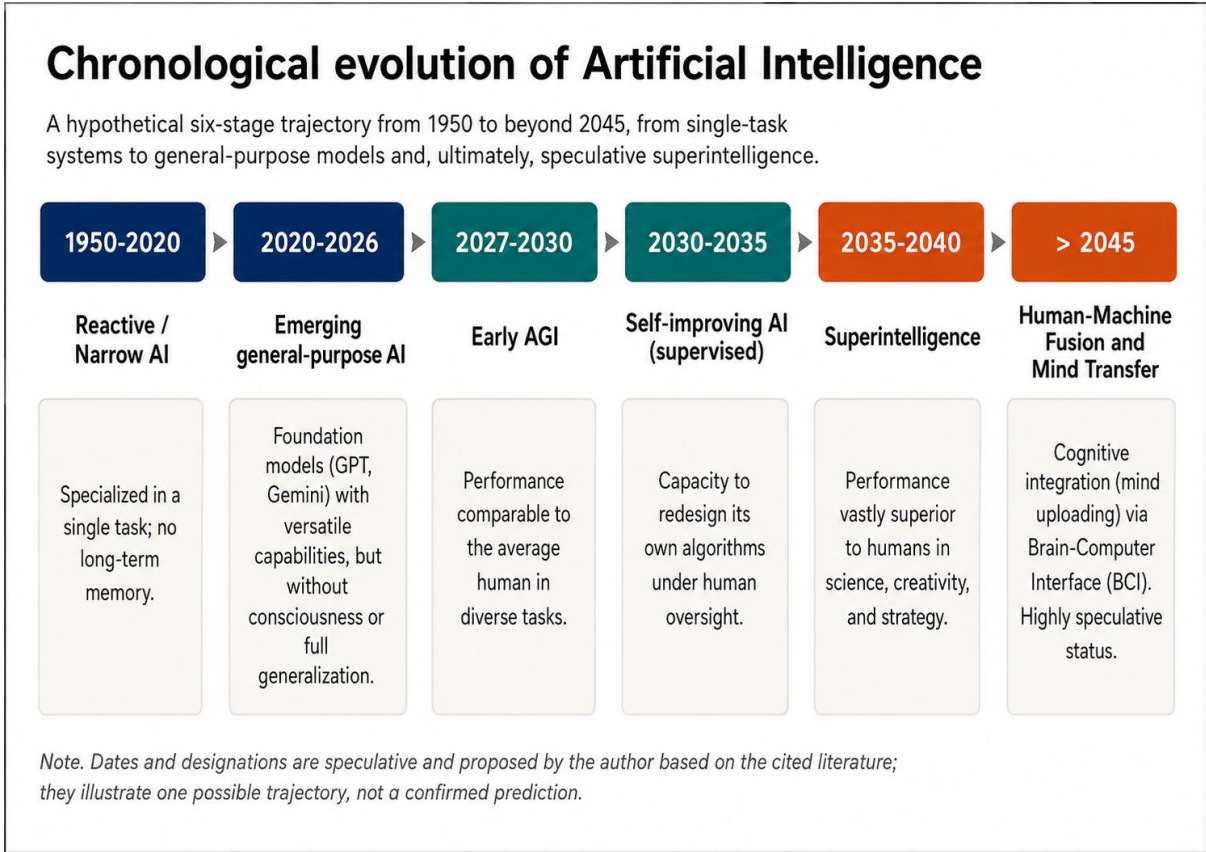


Figure 1. Infographic illustrating the hypothetical chronological evolution of AI.

3. Technological Singularity

The technological singularity is a hypothetical future scenario in which AI-driven growth becomes uncontrollable and irreversible (Vinge, 1993). This process would be driven by a recursive self-improvement cycle, in which an AI capable of redesigning itself would trigger an intelligence explosion, giving rise to a superintelligence with capabilities vastly superior to those of humans in virtually every domain.

Two fundamental conceptions are distinguished from this notion:

3.1 The Functional Singularity

This refers solely to the exponential growth of technical capabilities, without implying an ontological mutation in the human being. AI becomes extraordinarily capable, but does not acquire consciousness, a soul, or moral status; it remains an artifact. From the Thomistic framework, this variant does not pose an ontological problem: the development of increasingly powerful tools is consistent with human nature as a technical being, provided that such tools remain ordered toward the good of the person.

3.2 The Strong Ontological Singularity

A term proposed by the author (§1.1): it posits the transformation or substitution of the essential nature of the human being, either because a machine attains the status of a person (consciousness, intellect, soul, freedom, moral dignity) or because the human being becomes, in essence, an artifact.

The analysis of this paper will focus on the singularity in general and, in particular, on this strong ontological variant, which constitutes the most radical position in the debate.

4. Reference Framework: Thomism

4.1 What Is Thomism?



Figure 2. Saint Thomas Aquinas by the Spanish painter Bartolomé Esteban Murillo (1650).

Thomism is the philosophical and theological current that arises from the teachings of St. Thomas Aquinas (1225–1274). Aquinas incorporated the knowledge of Aristotle (ca. 350 BC)—considered the father of logic—and brought it to perfection, integrating it into a complete logical system in which ideas are linked like pieces of a puzzle, and each conclusion follows necessarily from the preceding ones. To understand it better: if a few basic principles are accepted (for example, that nothing arises from nothing, or that a thing cannot “be and not be” at the same time and in the same respect), then, by applying step-by-step reasoning, Thomism leads to solid conclusions about the physical universe, human nature, and the transcendent. This internal structure is what leads many to regard Thomism as a philosophical system of remarkable deductive coherence: once its premises are accepted, each conclusion follows necessarily from the preceding ones, in a manner analogous—allowing for differences in nature—to a mathematical proof. It is, in any case, a coherence internal to its own framework, which implies that its validity depends on the rational acceptance of its metaphysical starting points. Among its fundamental features:

- **Moderate realism:** universal concepts are derived from sensory experience; things exist independently of the one who thinks them.
- **Corollary of the principle of causality:** “none gives what it does not have”; an effect cannot possess a perfection not present in its cause (the principle of causality proper is formulated in §4.3).
- **Distinction between natural substance and artifact:** the former possesses an intrinsic principle of operation; the artifact has a form imposed from without.

4.2 What Does the Thomistic Framework Contribute to This Study?

Thomism offers four key advantages for analyzing AI:

- **It avoids technological reductionism.** It distinguishes between operations that a machine can imitate and intelligence as a faculty for understanding reality, making it possible to differentiate simulation from genuine understanding.
- **It defines what a human being is.** It characterizes the person as a substantial unity of matter and its vital principle (rational form or soul), identifying which aspects of humanity are emulable and which are ontologically irreducible.
- **It provides operative metaphysical categories.** Concepts such as potency/act, substance/accident, and the four types of causality allow for a structural analysis of the difference between the living and the artificial.
- **It provides a foundation for an ethics of AI.** It distinguishes between persons (subjects of rights) and artifacts (tools).

4.3 Fundamental Premises and Logical Principles

Thomistic analysis rests on the following premises:

- **Principle of Non-Contradiction:** it is impossible for the same thing to “be and not be” at the same time and in the same respect. A door cannot be open and closed simultaneously.
- **Principle of Identity:** every being is identical to itself. A cat is a cat; it cannot, at the same time, be a dog.
- **Principle of Sufficient Reason:** nothing occurs without a cause or reason to explain it. If a glass of water appears on the table, someone must have placed it there.
- **Principle of the Analogy of Being:** “being” is not predicated in the same way of all entities. We say that a person is intelligent, that a parrot is skilled at imitating sounds, and that a calculator is fast. These are entirely distinct modes of being.
- **Principle of Causality:** every “contingent being” (one that begins to exist or may not exist) has a cause that produces it.
- **Principle of Finality:** every “agent” acts for an end (seeking an objective, a good, or a perfection proper to it).
- **Distinction between Essence and Existence:** in all created beings, what something is (its essence) is distinct from the fact that it exists. We may have in mind a perfectly defined essence of a “unicorn,” but that does not mean that unicorns exist.
- **The Intellect as the Faculty of the Universal:** the human being possesses a capacity that neither animal nor machine possesses: the capacity to abstract from sensory data the universal essence of things. A child sees several different chairs and grasps the universal idea of “chair.” An AI, by contrast, can only compare the image with millions of previously labeled examples and calculate a probability; it does not understand what it means to sit.

4.4 Natural Substance and Artifact (The Key Distinction)

For Thomism, there are two fundamental types of beings:

- **Natural substance:** something that exists in itself and has within it the principle of its own activity. Examples: a dog, a tree, a human being. What makes something a natural substance is its substantial form (in the human being: the rational soul).
- **Artifact:** something fabricated by an external (human) intelligence. Its organization and functioning do not arise from within, but are imposed from without by its designer. Examples: a chair, a robot, an AI program.

Categorical (not gradual) difference: this is not a matter of the natural substance having “more” life and the artifact “less.” Rather, the mode of being is radically distinct. An AI program, however complex, remains an artifact: its “intelligence” is a functional imitation, not an internal principle of genuine understanding.

4.5 Potency and Act

- **Act:** what something is now (its current state).
- **Potency:** the capacity to become something it is not yet. A seed is, in act, a seed; but it has the potency to become a tree.

Application to AI: a calculator has the potency (capacity) to add numbers, but that potency does not arise from itself: it was given by its creators in the design of the circuit. AI has no potencies of its own; it only actualizes the functions programmed into it.

4.6 The Four Causes

Table 2 illustrates the four Aristotelian causes applied to artificial intelligence (Aristotle, ca. 350 BC).

Table 2. *The Four Causes Applied to AI*

Cause	Question	Example Applied to AI
Material	What is it made of?	Chips, cables, electricity, servers.
Formal	What is its structure or essence?	The code, the algorithms, the software architecture.
Efficient	Who made it?	The engineers, programmers, and companies that designed it.
Final	What is it for?	To assist, automate, recommend, predict (ends assigned by humans).

Note. The final cause (purpose) is the most important. In AI, purpose does not arise from within itself; it is given by its creators.

4.7 The Soul as Vital Principle

When Thomism speaks of the “soul,” it does not refer to a ghost or a mysterious energy. It refers to the internal principle that organizes a living being and causes it to act from within (Aquinas, n.d.). Three types are distinguished:

- **Vegetative Soul:** the most basic, present in all living beings. Its principal functions are nutrition, growth, and reproduction.
- **Sensitive Soul:** proper to animals and human beings. It adds the capacities of sensory perception, local movement, sensitive appetites, and memory and imagination.
- **Rational Soul:** exclusive to the human being. It includes the capacities of the vegetative and sensitive souls and adds intellectual operations: understanding, reasoning, learning universal essences, and exercising will and freedom.

In human beings, these are not three separate souls. We possess a single, unique vital principle: the rational soul, which contains and exercises in unity all the functions proper to the vegetative and sensitive levels.

4.8 Human Cognitive Process According to Thomism

St. Thomas Aquinas (n.d.) described human knowledge as a process that begins in the senses and culminates in the intellect. Contemporary neuroscience distinguishes functionally equivalent stages: sensory information is processed in the primary cortex, integrated in association areas (parietal and temporal cortex), categorized into patterns, and finally consolidated into concepts that guide reasoning in the prefrontal cortex. Aquinas’s model, formulated more than 700 years ago, remains a valid framework for describing this process (Ocampo Ponce, 2024; Martínez Barrera, 2011).

Table 3 summarizes the faculties of the human cognitive process.

Table 3. Faculties of the Human Cognitive Process

Cognitive Faculty	Function	What It Does
External Senses	Apprehend concrete qualities	Perceive color, shape, texture, etc.
Common Sense	Unify data from the different senses	Recognize that what is brown, rough, and makes a sound is the same object.
Imagination	Retain concrete images	Store the “mental picture” of that individual object.
Cogitative Power	Perceive practical utility or meaning	“This seems useful for sitting” (prior to concept formation).
Agent Intellect	Abstract the universal essence from the concrete image	Disregard details (color, material) and grasp the “what it is.”
Possible Intellect	Receive, understand, and retain the abstract concept	Keep the idea of “chair” permanently available.

Note. Adapted from Aquinas (n.d.) and contemporary neuroscience.

An Example: Knowing a Chair

Consider the case of a child encountering a chair for the first time, and compare it with an AI system's simulation of that process.

What Does the Human Being Do?

- **External senses:** perceive something brown, with four legs, a flat surface, and a backrest. Touch its smooth texture.
- **Common sense:** the mind unifies all those sensations and communicates: "all of this is the same object."
- **Imagination:** retains a mental image of that particular chair (mahogany color, high backrest, etc.).
- **Cogitative power:** without yet possessing the concept, intuits that the object "seems good for sitting." This is an intelligent perception, not an instinctive one.
- **Agent intellect:** the intelligence "illuminates" the image, separates the accidental details (wood, brown) and grasps the universal essence: an object with a horizontal surface and a backrest, designed to be sat upon.
- **Possible intellect:** receives that essence and forms the stable concept of chair. From that moment, the child will recognize any chair for the rest of its life.

What Does AI Do?

AI possesses none of these faculties in the proper sense:

- It replaces the senses with sensors (cameras, microphones, etc.): it captures data.
- It uses data-fusion algorithms to unify information (analogous to common sense, but without consciousness).
- It stores statistical patterns instead of mental images.
- It has no cogitative power: it does not intuit utility or meaning; it only calculates probabilities.
- It has no agent intellect: it does not abstract essences; it only finds correlations between data.
- It has no possible intellect: it does not form universal concepts with meaning; it only assigns probabilistic labels.

Corollary: the human being sees a new chair and understands that it is used for sitting. AI compares the image with millions of examples labeled "chair" and responds "98% probability that this is a chair." It does not know what sitting means. This comparison is revisited and formalized in Chapter 7, where the full ontological distinction between the human being and AI is established.

5. Contributions of Contemporary Science to the Thomistic Framework

Thomistic metaphysics is not a closed system; it can engage in dialogue with the most recent scientific discoveries. This chapter presents two contributions from contemporary science that reinforce the theoretical framework of this paper: the contemporary cosmological framework—which confirms the contingency of the universe and its intelligible order—and Gödel’s incompleteness theorem, which demonstrates the intrinsic limits of every formal system, including AI.

5.1 The Contemporary Cosmological Framework and Thomistic Metaphysical Realism

Two findings from twentieth-century physics and cosmology are especially relevant to the Thomistic framework. The first is the Big Bang theory—postulated by Georges Lemaître in 1931 (Lemaître, 1931) and today widely accepted—which establishes that the universe had a definite beginning approximately 13.8 billion years ago: matter and time likewise had a beginning (Hawking, 1988). If the universe began to exist, the Thomistic principle of causality invites the question of its cause, which cannot itself be physical. The second finding is the so-called “fine-tuning” of the physical constants: a set of fixed numbers—such as the strength of gravity or the mass of the electron—which, if altered by even a minimal fraction, would make impossible the existence of stars, chemistry, or life (Rees, 1999). This extraordinarily precise order actualizes Aquinas’s Fifth Way: the constant orientation of natural things toward their ends points to an ordering Intelligence.

It should be noted that the move from scientific data to philosophical conclusion requires additional argumentative steps—alternative positions exist, such as quantum models without an initial singularity or the multiverse hypothesis. However, the contribution of these findings to the framework of this paper is more circumscribed and precise: they confirm that the universe is contingent (it does not explain itself) and that it exhibits an intelligible order, two premises that Thomism already maintains and that contemporary science supports empirically. The central argument of the following section rests on this ontological foundation.

5.2 Gödel’s Theorem and the Limits of Formal Systems

Kurt Gödel, the Austrian mathematician and contemporary of Albert Einstein, published in 1931 a theorem that many regard as one of the most important intellectual milestones of the twentieth century (Gödel, 1931; Raatikainen, 2014).

What Is a “Formal System”?

In mathematics, a formal system is a set of clear and precise rules that allow theorems to be generated mechanically, step by step, without the need for interpretation or meaning. One begins with assumed truths (axioms) and applies mechanical rules to deduce new information. This makes it possible to study the structure of pure reasoning, independently of what the symbols mean.

A computer—including any AI system—can be understood, in this relevant sense, as a formal system: it executes rules programmed by its creators and manipulates symbols according to predefined algorithms.

Gödel's Theorem: The Dream of Completeness Collapses

In the early twentieth century, the great mathematician David Hilbert dreamed of constructing a perfect mathematical system: one that would be both complete and consistent. Gödel (1931) demonstrated that this was impossible. His incompleteness theorems establish two fundamental propositions:

- **First theorem:** in any sufficiently complex formal system, there exist true statements that cannot be proven within the system itself. That is, the system is intrinsically incomplete.
- **Second theorem:** that same system cannot prove its own consistency using only its own rules.

Gödel constructed a statement that said, in essence: “This assertion cannot be proven within this system.” If the system were to prove the statement true, it would be false (because it asserts that it is unprovable). If the system were to prove it false, it would be true. The only way out is to accept that the statement is true but unprovable within the system.

AI as a Formal System: An Inherited Incompleteness

Computation—and therefore any artificial intelligence system—inherits this character of a formal system: it operates by means of algorithms, that is, finite sets of rules applied mechanically. Consequently, everything that an AI can “know” or “prove” is limited by the rules that its creators have programmed into it. In this sense, Gödel’s theorem provides a weighty contribution: insofar as artificial intelligence operates as a formal system, it is subject to intrinsic limits that it cannot overcome from within its own rules. No formal system of this kind can guarantee its own internal consistency or prove all the truths expressible in its own language (Gödel, 1931; Raatikainen, 2014); for the implications of this limitation for artificial intelligence in particular, see also Observatorio de Bioética (2025) and Rivero González (2025).

It should be noted that this identification of AI with a formal system is more straightforward in the case of classical symbolic systems, based on explicit logical rules, than in the case of contemporary deep learning systems, whose behavior emerges from the statistical adjustment of parameters rather than from an explicit axiomatics. Even so, since these systems ultimately run on deterministic digital hardware—governed by a finite set of discrete logical-arithmetic operations—they retain in a broad sense the character of a formal system relevant to the argument, although the mapping between “rules” and “trained weights” is less transparent than in a traditional axiomatic system. This qualification does not weaken the central conclusion of this section, but requires that it be advanced with the same argumentative caution applied below in examining the Penrose-Lucas argument.

The Penrose-Lucas Argument: The Human Mind and Algorithms

The philosopher John Lucas (1961) applied Gödel’s theorem to the debate on artificial intelligence, an argument that the mathematician and Nobel Prize in Physics Roger Penrose took up and reformulated independently almost three decades later (Penrose, 1989/1996), giving rise to what is now known as the Penrose-Lucas argument. This argument holds the following:

- A machine (as a formal system) cannot prove the truth of its own “Gödel statement.”
- Yet a human mathematician can recognize, from outside the system, that the statement is true, even though the system cannot prove it.
- Therefore, the human mind cannot be identical to a computational machine: there is something in human understanding that transcends algorithms.

It is important to note that this argument is one of the most influential in the philosophy of mind, but also one of the most academically contested. Among the most frequently cited objections: Benacerraf (1967) argued that, for the argument to work, it must first be possible to explicitly construct the Gödel sentence of the formal system attributed to the human mind; if that system is too complex to characterize with precision, such a construction might not be possible, and the argument would be unable to get off the ground. LaForte, Hayes, and Ford (1998), for their part, objected not on grounds of a *petitio principii*, but rather identified a terminological ambiguity: they argued that Penrose's reasoning shifts, without acknowledging it, between precise and informal senses of key terms such as 'proof' and 'truth,' and that this shift is what generates the appearance of an anti-mechanist conclusion. From the Thomistic framework adopted in this paper, however, the Penrose-Lucas argument is coherent with the distinction between the agent intellect and algorithm established in the preceding sections, a distinction grounded independently in Thomistic metaphysics.

"The conclusion seems inescapable: mathematicians are not using a knowable and verifiable computational procedure to determine mathematical truth; it follows that mathematical understanding cannot be reduced to unconscious computation" (Penrose, 1989/1996).

It is necessary to clarify, with academic rigor, that Gödel's theorem does not by itself prove that the human mind transcends formal systems: establishing this would first require proving that the mind is not a formal system, which is precisely the point at issue. What the theorem does permit affirming, in combination with Thomistic premises, is the following: if the human mind possesses a genuine faculty of abstraction—the agent intellect—that is not reducible to algorithmic operations, then that faculty is situated, by definition, beyond the reach of any formal system. In this sense, Gödel's theorem functions as a logical confirmation of the limits of the artificial, while the distinction between intellect and algorithm rests on the Thomistic philosophical foundation.

6. Contributions of the Encyclical Magnifica Humanitas (Leo XIV, 2026)

6.1 The Document and Its Relevance to This Paper



Figure 3. Pope Leo XIV with Christopher Olah (co-founder of Anthropic) and Professor Anna Rowlands (University of Durham) at the presentation of the Encyclical Letter *Magnifica Humanitas*, May 2026.

Pope Leo XIV signed the Encyclical *Magnifica Humanitas* on 15 May 2026, with the subtitle “On the Safeguarding of the Human Person in the Age of Artificial Intelligence.” It constitutes the first document of the Magisterium dedicated entirely to the impact of AI on human dignity and the social doctrine of the Church (Leo XIV, 2026). Its contribution to the framework of this paper is threefold: it confirms, from magisterial authority, the ontological distinction between person and artifact that Thomism grounds philosophically; it denounces the technocratic paradigm that underlies the singularity hypothesis; and it proposes ethical criteria for orienting AI development toward the common good.

6.2 AI as Artifact: Convergence with the Thomistic Framework

The Encyclical defines AI as a system that imitates certain human cognitive functions, but does not possess genuine understanding or subjectivity: “these systems merely imitate certain functions of human intelligence,” yet they “do not undergo experiences, do not possess a body, do not feel joy or pain,” nor do they “have a moral conscience,” since they do not judge good and evil (Leo XIV, 2026, n. 99). Pope Leo XIV insists that AI lacks a body, affections, moral experience, and consciousness: it can simulate empathy, but cannot feel it (n. 99). This characterization is fully consistent with the Thomistic distinction between natural substance and artifact developed in Chapter 4: AI does not possess an internal vital principle, but a form imposed from without by its designers. Far from being neutral, it “takes on the characteristics of those who devise, finance, regulate and use it” (n. 9).

At this point, the Encyclical may be read in convergence with the position advanced in this paper: the sharp distinction the document draws between human intelligence and artificial intelligence is consistent with the rejection, from Thomistic philosophy, of the strong ontological singularity—the hypothesis that an AI could become a person or a human being could become an artifact. The artificial imitation of communication can induce deception, especially in lonely or vulnerable persons (n. 100), precisely because superficial resemblance conceals a fundamental ontological difference.

Three principal risks of the personal use of AI (n. 100):

- **Ease of delegation:** the speed of AI can weaken personal judgment and creativity.
- **Illusion of objectivity:** responses appear neutral, but reflect the cultural parameters and biases of their designers.
- **Simulation of human relationship:** the imitation of empathy can create fictitious bonds and substitute the desire for genuine encounter.

6.3 Rejection of the Technocratic Paradigm and the Strong Ontological Singularity

The Encyclical explicitly analyzes the transhumanist and posthumanist currents (nn. 115–118), which constitute the ideological substrate of the strong ontological singularity as defined in this paper. Leo XIV does not reject technology itself, but rather the anthropological vision that animates these currents: the idea that the human limit—illness, old age, frailty, death—is a correctable error through technology. Confronting this paradigm, the Pope affirms: “humanity flourishes not despite limitations, but often through them” (n. 118). Frailty is not a design flaw; it is the space in which compassion, tenderness, and solidarity are born. This magisterial position converges directly with the central conclusion of this paper: the strong ontological singularity is not only philosophically incoherent according to Thomism, but ethically inadmissible according to the social doctrine of the Church, because it entails reducing the human being to an improvable system or a replaceable substrate.

6.4 The Social Challenge: Hyperautomation, Work, and Dignity

The Encyclical illuminates with particular acuity the true challenge posed in the conclusion of this paper: not whether machines will eventually awaken, but what place the human being will occupy in a world of hyperautomation. Leo XIV denounces that “innovation is often pursued solely for reducing costs and increasing profits” (n. 151), and that behind the apparent immateriality of AI there are concrete chains of exploitation—the so-called invisible labor: millions of people who label data and train models for meager wages, mineral extraction under inhumane conditions, and the mass control of data about fragile populations as a new form of data colonialism (nn. 173–178). In response, the document demands traceability of algorithmic decisions, effective human oversight, protection of data as a common good, and coins the expression “disarming AI”: withdrawing it from the logic of the arms race—today not only military, but also economic and cognitive—breaking the equivalence between technological power and the right to govern, and freeing it from monopolies in order to orient it toward the good of all humanity (n. 110).

6.5 Corollary: What the Encyclical Contributes to the Central Argument

For this paper, Magnifica Humanitas provides three elements that reinforce and complete the Thomistic framework: (1) magisterial confirmation of the ontological distinction between person (natural substance with a rational soul) and artifact (AI), operating on the ethical and social plane what Thomism grounds philosophically; (2) denunciation of the technocratic paradigm underlying the singularity hypothesis, which reduces the person to a cog; and (3) a positive proposal: an AI in the service of the human being, oriented toward the common good, in which every person has a place, dignified work, truth, and freedom. This horizon coincides with the “third way” proposed in the conclusion of this paper.

7. Human Being and AI: Ontological Distinction

7.1 What Is the Human Being According to Thomism?

The human being is a natural substance: it exists in itself and possesses a vital principle (rational soul) that grants it genuine intelligence, will, and self-consciousness (Aquinas, n.d.; Martínez Barrera, 2011).

7.2 What Is AI According to Thomism?

AI is an artifact: its form is imposed from without. It lacks an intrinsic principle of operation, life in the philosophical sense, consciousness, and intellect (Ocampo Ponce, 2024).

7.3 Key Differences

Table 4 summarizes the fundamental differences between the human being and AI from the Thomistic perspective.

Table 4. Key Differences Between the Human Being and AI

Aspect	Human Being	AI
Basis of knowledge	Abstracts universal essences	Correlates data and statistical patterns
Genuine understanding	Yes (grasps meanings, intentions)	No (processes symbols without understanding)
Consciousness	Yes (subjective experience)	No
Reflexivity	Yes (can analyze its own thoughts)	No (only executes algorithms)

Note. Adapted from Aquinas (n.d.) and Leo XIV (2026).

7.4 Levels of Human-AI Interaction

Table 5 organizes the possible modes of integration between the human being and AI, from lesser to greater impact on human nature.

Table 5. Levels of Human-AI Interaction (from Lesser to Greater Impact)

Level	Type of Interaction	Example	Does It Alter Human Nature?
1	Instrumental use	Using a calculator or ChatGPT	No (external tool)
2	External prosthesis	Glasses, hearing aids, prosthetic arm	No (replace lost functions)
3	Reversible functional implant	Pacemaker, cochlear implant	No (assist bodily functions without altering personal identity)
4	Partial replacement of brain tissue	Hypothetical neuromorphic memory chip	Borderline (could affect identity)
5	Total brain replacement	Mind uploading, BCI, and avatars	Yes (would cease to be human according to Thomism)

7.5 Formalization of the “Strong Ontological Singularity”

We define two sets: S = the set of natural substances (human beings, animals, plants) and A = the set of artifacts (chairs, robots, AI programs). According to the Thomistic framework, these two categories are mutually exclusive: $S \cap A = \emptyset$. This means that no element is simultaneously a natural substance and an artifact.

Table 6 shows the transformations implied by the strong ontological singularity and the response of each philosophical framework.

Table 6. Possible Transformations of the Strong Ontological Singularity

Transformation	Meaning	Computationalist / Transhumanist View	Thomistic View (this paper)
Human → Artifact	The mind becomes an executable program in another substrate	Possible (the mind is only information)	Not coherent: the substantial form (soul) cannot bestow its being upon artificial matter.
AI → Natural Substance	An AI acquires genuine consciousness, understanding, and will	Possible (emergence from complexity)	Not coherent: violates the principle of causality.

Note. Adapted from Aquinas (n.d.) and Bostrom (2014).

7.6 Corollary: The Strong Ontological Singularity Is Incoherent

Within the Thomistic framework, the hypothesis that an AI could become a person, or that a human being could become an artifact, contradicts two basic principles: categorical exclusion and the principle of causality.

“Artificial intelligence can imitate human functions with increasing sophistication, but this does not imply ontological equivalence with the human person. The difference is not one of degree (more or less complexity), but of mode of being: a difference of nature—internal principle of operation—not merely of function executed.” (Author’s own elaboration)

Human beings and AI: an ontological distinction

From the Thomistic perspective, human beings and AI belong to distinct and irreducible categories of being. Key differences, levels of interaction, and possible transformations of the strong ontological singularity.

1 Key differences

between human beings and AI

Aspect	Human Being	AI
Basis of knowledge	Abstracts universal essences	Correlates data and statistical patterns
Genuine understanding	Yes (grasps meanings, intentions)	No (processes symbols without understanding)
Consciousness	Yes (subjective experience)	No
Reflexivity	Yes (can analyze its own thoughts)	No (only executes algorithms)

The difference is not a matter of degree (more or less complexity), but of being.

2 Levels of human-AI interaction

(from lower to greater impact)

1	Instrumental use Using a calculator or ChatGPT	No (external tool)
2	External prosthesis Glasses, hearing aids, prosthetic arm	No (replace lost functions)
3	Reversible functional implant Pacemaker, cochlear implant	No (assist bodily functions without altering personal identity)
4	Partial replacement of brain tissue Hypothetical neuromorphic memory chip	Borderline (could affect identity)
5	Total replacement of the brain Mind uploading, BCI, and avatars	Yes (would cease to be human according to Thomism)

3 Possible transformations

of strong ontological singularity

	Meaning	Computationalist / Transhumanist View	Thomistic view (this paper)
A Human → Artifact	The mind becomes an executable program in another substrate	Possible (the mind is only information)	Not coherent: the substantial form (soul) cannot bestow its being upon artificial matter.
B AI → Natural substance	An AI acquires genuine consciousness, understanding, and will	Possible (emergence from complexity)	Not coherent: violates the principle of causality.

Corollary: strong ontological singularity is incoherent —

Within the Thomistic framework, the hypothesis that an AI becomes a person or a human becomes an artifact contradicts two basic principles: categorical exclusion and the principle of causality.

"Artificial intelligence can imitate human functions with increasing sophistication, but this does not imply ontological equivalence with the human person."

Figure 4. Infographic illustrating the ontological distinction between human beings and AI.

8. General Conclusion

Restatement of the Question

This paper began by asking whether a machine can ever become a person. The answer, within the Thomistic framework adopted, is clear: no. But this is not a simple negation; it is an affirmation grounded in logic and in the nature of the universe.

Synthesis of the Argument

We have seen that artificial intelligence, however sophisticated, is nothing more than an artifact: its rules, ends, and functioning are imposed from without by its creators. Thomism has provided us with a solid logical framework for distinguishing between what acts from an internal principle (natural substance, such as the human being) and what merely executes what has been programmed (artifact, such as AI). This difference follows from the fact that their natures are logically mutually exclusive: a thing cannot be both at once.

Contemporary science reinforces this distinction. The current cosmological framework—the Big Bang and the fine-tuning of the physical constants—confirms that the universe is contingent and exhibits an intelligible order that does not explain itself, two premises that Thomism already upholds and that twentieth-century physics has supported empirically. And Gödel's theorem (1931) demonstrates that no formal system can exhaust truth from within its own rules; combined with Thomistic premises, it points to the fact that the human mind operates in a register that algorithms cannot attain, though it is important to underline that this conclusion rests on the Thomistic philosophical premise and not solely on the mathematical theorem.

The Encyclical Magnifica Humanitas (Leo XIV, 2026) confirms from the Magisterium that AI is not neutral, that it can imitate the human, but does not possess consciousness or subjectivity, and denounces the technocratic paradigm that reduces the person to a cog.

Final Answer

For all these reasons, the most radical hypothesis within the AI singularity scenario—the so-called strong ontological singularity (a term proposed by the author), according to which an AI would become a person or a human being would become an artifact—is incoherent with the logical and causal principles set forth above. A machine will never be a person, because it lacks the vital principle we call the soul.

Final Reflection

The true challenge, however, is not whether machines will eventually awaken. The challenge is what place we, human beings, will occupy in an increasingly automated world, where efficiency and economic profit threaten to reduce us to data, to profiles, to dispensable cogs.

Faced with two extremes—the transhumanist fusion that dissolves the distinction between person and artifact, and the technocratic paradigm that reduces the person to a cog serving profit—this paper aligns with a third way: that of an artificial intelligence in the service of the human being. A technology that empowers persons without annulling them, that respects their limits without seeking to eliminate them, that protects their freedoms rather than dissolving them in algorithms governed by others.

The most real singularity will not be that of the machines, but that of a future in which technology—recognized as a product of human ingenuity—remains in the service of persons, and not the reverse.

That future is possible. And it begins today, with every decision we make as a society, as a scientific community, as entrepreneurs, as policymakers, and above all as persons who still ask themselves:

Does your virtual assistant feel anything when it converses with you? The answer remains: no. But the question itself is profoundly human—and it has been so since the origin of humanity.

9. Glossary

9.1 Thomistic — General

Term	Meaning
Abstraction	Mental operation that separates the universal essence from concrete details in order to form a general concept.
Accident	Quality or part added to a substance that does not alter its essence (e.g., a dental implant, hair color).
Soul	Internal principle that organizes a living being and causes it to act from within.
Artifact	Something fabricated by an external (human) intelligence, whose organization is imposed from without (e.g., a chair, a robot, an AI program).
Cause (four types)	Modes of answering why something is: material (what it is made of), formal (its structure), efficient (who made it), final (what it is for).
Contingent	Something that can exist or not exist; its existence depends on an external cause. The universe is contingent (it had a beginning; it could not have been).
Essence / Existence	Essence is what a thing is (its nature or set of defining characteristics). It is that which allows the object to be defined. Existence is the act by which that thing is real and present in the world.
Substantial Form	Internal principle that makes a substance what it is (in the human being: the rational soul).
Potency / Act	Potency: the capacity to become something not yet actual. Act: the state of already being it.
Principle of the Analogy of Being	“Being” is not predicated identically (univocally) nor entirely differently (equivocally) of all entities, but proportionally.
Principle of Causality	Postulates: “Every contingent being (one that begins to exist or may not exist) has a cause that produces it.”
Principle of Finality	Postulates: “Every agent acts for an end.” Any being that acts, operates, or moves in nature does so in pursuit of an objective, a good, or a perfection proper to it.
Principle of Identity	Postulates: “Every being is identical to itself.”
Principle of Non-Contradiction	A thing cannot “be and not be” at the same time and in the same respect.
Principle of Sufficient Reason	Postulates: “Every being (or proposition) has a reason for being.” If it exists by itself, its reason lies in its own essence; if by another, its reason lies in an external cause.
Natural Substance	Something that exists in itself and acts from within (e.g., a human being, a tree, a dog).

9.2 Thomistic — Cognitive Process

Term	Meaning
Cogitative Power	Intelligent perception of practical utility or meaning. This faculty is exclusive to the human being as a rational being.
Universal Concept	General idea (e.g., “chair,” “justice,” “beauty”) applicable to many particular cases.
Imagination	Faculty that retains concrete images of individual objects.
Intellect	Human faculty of understanding, grasping the essence of things, and forming universal concepts.
Agent Intellect	Faculty that abstracts the universal essence from sensory data.
Possible Intellect	Faculty that receives and retains the abstract concept, forming stable universal ideas.
Common Sense	Capacity to unify data from the different senses into a single object.
External Senses	Sight, hearing, touch, taste, smell. They apprehend concrete qualities.

9.3 Encyclical Magnifica Humanitas (Leo XIV, 2026)

Term	Meaning
Data Colonialism	The mass control of data by large corporations constitutes a new form of colonialism.
Technocratic Paradigm	The tendency to allow the logic of efficiency, control, and profit to govern all human decisions, reducing the person to a cog.
Posthumanism	Current that critiques anthropocentrism and proposes total hybridization of human, machine, and environment.
Invisible Labor	Unrecognized human work that sustains AI systems (data labeling, content moderation, mineral extraction), often under exploitative conditions.
Transhumanism	Current that aspires to “enhance” the human being through technology (implants, algorithms, fusion with machines).

9.4 Miscellaneous

Term	Meaning
Avatar	Artificial (robotic or other synthetic) support or body to which human consciousness or mind would be transferred, allowing it to interact with the real world.
Brain-Computer Interface (BCI)	Technological system that establishes a direct communication pathway between the brain and an external device. Currently in an experimental phase.
Neuromorphic Chip	Type of processor designed to mimic the neuronal architecture of the human brain in order to make computation more efficient.
Emergence	Phenomenon by which a property or entity appears spontaneously from simpler interactions, without explicit programming.
Metaphysics	Philosophical discipline that studies being as being, i.e., the first causes and principles of reality.
Mind Uploading	Hypothetical process of completely emulating the human mind in a digital substrate.
Ontology	Branch of metaphysics concerned with the study of being in general and its properties. In Thomism, ontology and metaphysics are practically identified.
Strong Ontological Singularity	Term proposed by the author to designate the most radical variant of the singularity scenario: that in which an AI becomes a person or a human being becomes an artifact.
Functional Singularity	Variant of the singularity scenario referring solely to the exponential growth of AI's technical capabilities, without implying any change in the nature of the human being.
Formal System	An abstract and logical structure used to prove theorems rigorously. It consists of four elements: Formal language: A finite set of symbols combined to form formulas or mathematical expressions. Axioms: A set of formulas accepted as true by initial definition, without the need for proof within the system. Rules of inference: The 'engine' of the system. These are the logical rules that allow one to take axioms or other previously proven formulas and transform or combine them to derive new formulas. Theorems: The formulas successfully constructed step-by-step from the axioms through the application of the rules of inference.
Penrose-Lucas Argument	Argument formulated by the philosopher John Lucas (1961) and the mathematician Roger Penrose (1989/1996) from Gödel's theorem: a machine cannot prove the truth of its own "Gödel statement," but a human mathematician can recognize it from outside the system. Therefore, the human mind cannot be reduced to an algorithm.
Functionalism / Computationalism	Functionalism (Putnam, Fodor) holds that the mind can be reproduced in any substrate that performs the same cognitive functions. Computationalism (Turing, Minsky) proposes that thought is, in essence, information processing.
Fine-Tuning of Physical Constants	Cosmological phenomenon by which the physical constants of the universe (such as the strength of gravity or the mass of the electron) have extremely precise values: a minimal variation in any of them would make impossible the existence of stars, chemistry, or life.

10. List of Tables and Figures

10.1 List of Tables

Table	Title
Table 1	Hypothetical Chronological Evolution of AI
Table 2	The Four Causes Applied to AI
Table 3	Faculties of the Human Cognitive Process
Table 4	Key Differences Between the Human Being and AI
Table 5	Levels of Human-AI Interaction (from Lesser to Greater Impact)
Table 6	Possible Transformations of the Strong Ontological Singularity

10.2 List of Figures

Figure	Title / Description
Cover	“Dieste.” Tribute to the work of Engineer Eladio Dieste (Church of Atlántida – Canelones, Uruguay. UNESCO World Heritage Site). Figure created with original photographs by the author and artistically composed with AI. Author: Roberto Asplanato. June 2026.
Figure 1	“Infographic of the Hypothetical Chronological Evolution of AI.” Original figure created by the author based on Table 1 of this paper. Author: Roberto Asplanato. June 2026.
Figure 2	“St. Thomas Aquinas” by Bartolomé Esteban Murillo (Spanish painter). Own work, Amuley, CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=33176874
Figure 3	Pope Leo XIV alongside Christopher Olah and Professor Anna Rowlands [Video still]. 2026. Source: Cernuzio, S. (2026, May 25). Leo XIV presents the Encyclical: disarm AI, reject logics of exclusion and domination [Video]. Vatican News. https://www.vaticannews.va/es/papa/news/2026-05/papa-leonpresentacion-enciclica-magnifica-humanitas.html
Figure 4	“Ontological Distinction Between the Human Being and AI.” Original figure created by the author based on Chapter 7 of this paper (Tables 4 and 6). Author: Roberto Asplanato. June 2026.

Acknowledgements

The author would like to first express gratitude to the Blessed Virgin Mary and Saint Thomas Aquinas, who provided the inspiration and spiritual guidance for this work. Special thanks are also extended to Valeria and Iván for their constant support and companionship throughout the preparation of this article.

11. References

- Altman, S. (2023). Planning for AGI and beyond. OpenAI Blog. <https://openai.com/index/planning-for-agi-and-beyond/>
- Aristotle. (ca. 350 BC/1994). *Metaphysics* (T. Calvo Martínez, Trans.). Editorial Gredos.
- Benacerraf, P. (1967). God, the devil, and Gödel. *The Monist*, 51(1), 9–32.
- Bostrom, N. (2014). *Superintelligence: Paths, dangers, strategies*. Oxford University Press.
- Fodor, J. A. (1975). *The Language of Thought*. Thomas Y. Crowell.
- Gödel, K. (1931). Über formal unentscheidbare Sätze der Principia Mathematica und verwandter Systeme I. *Monatshefte für Mathematik und Physik*, 38(1), 173–198.
- Hawking, S. (1988). *A brief history of time: From the Big Bang to black holes*. Bantam Books.
- Kurzweil, R. (2005). *The singularity is near: When humans transcend biology*. Viking Press.
- LaForte, G., Hayes, P. J., & Ford, K. M. (1998). Why Gödel's theorem cannot refute computationalism. *Artificial Intelligence*, 104(1–2), 265–286.
- Lemaître, G. (1931). The beginning of the world from the point of view of quantum theory. *Nature*, 127, 706.
- Leo XIV, Pope. (2026, May 15). Magnifica Humanitas: On the safeguarding of the human person in the time of artificial intelligence. Libreria Editrice Vaticana. <https://www.vatican.va/content/leo-xiv/en/encyclicals/documents/20260515-magnifica-humanitas.html>
- Lucas, J. R. (1961). Minds, machines, and Gödel. *Philosophy*, 36(137), 112–127.
- Martínez Barrera, J. (2011). Alma, cuerpo y mente: Santo Tomás y algunos contemporáneos [Soul, body, and mind: St. Thomas Aquinas and some contemporaries]. *Sapientia*, LXVII(229–230). <https://repositorio.uca.edu.ar/bitstream/123456789/4640/1/alma-cuerpo-mente-santo-tomas.pdf>
- Minsky, M. (1967). *Computation: Finite and infinite machines*. Prentice-Hall.
- Observatorio de Bioética. (2025). Gödel: un teorema que cambió nuestra visión del universo [Gödel: a theorem that changed our vision of the universe]. Universidad Católica de Valencia. <https://www.observatoriobioetica.org/2025/12/godel-un-teorema-que-cambio-nuestra-vision-del-universo/10006424>
- Ocampo Ponce, M. (2024). Inteligencia artificial y producción de conceptos en Tomás de Aquino [Artificial intelligence and concept production in Saint Thomas Aquinas]. *Revista Conocimiento y Acción*, 4(2). <https://doi.org/10.21555/cya.v4.i2.3212>
- Penrose, R. (1989/1996). *The emperor's new mind: Concerning computers, minds, and the laws of physics / La mente nueva del Emperador* (J. J. García Sanz, Trans.). Oxford University Press / Fondo de Cultura Económica.
- Putnam, H. (1967). Psychological predicates. In W. H. Capitan & D. D. Merrill (Eds.), *Art, mind, and religion* (pp. 37–48). University of Pittsburgh Press.

- Raatikainen, P. (2014). Gödel's incompleteness theorems. In E. N. Zalta (Ed.), *The Stanford encyclopedia of philosophy* (Spring 2014 ed.). Stanford University.
<https://plato.stanford.edu/archives/spr2014/entries/goedel-incompleteness/>
- Rees, M. (1999). *Just six numbers: The deep forces that shape the universe*. Basic Books.
- Rivero González, J. P. (2025). Del silicio al espíritu [From silicon to spirit], *journal of the Las Palmas Theological Center*, (76), 15–37.
<https://portalciencia.ull.es/documentos/6941ae342892130bab4389e5>
- Russell, S., & Norvig, P. (2020). *Artificial intelligence: A modern approach* (4th ed.). Pearson.
- Thomas Aquinas, St. (n.d.). *Summa Theologiae* (Blackfriars, Trans.). (Original work published ca. 1265–1274).
- Turing, A. (1950). Computing machinery and intelligence. *Mind*, 59(236), 433–460.
- Vinge, V. (1993). The coming technological singularity: How to survive in the post-human era. In *Vision-21: Interdisciplinary science and engineering in the era of cyberspace* (pp. 11–22). NASA Lewis Research Center.

Author's Note

This paper was conceived for any person interested in the topic, without requiring prior technical or philosophical training. Its purpose is to offer a rigorous, clear, and respectful reflection on human dignity. The author invites readers to maintain a critical mindset, to resist being swept along by currents contrary to our own nature, and to commit actively to building a just and genuinely human technological future.

ORCID ID: 0000-0002-3234-8680 | roberto.asplanato@outlook.com | June 2026